

Pattern Recognition (CSE4213)

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□ Linear Discriminant Analysis (LDA)

Outline

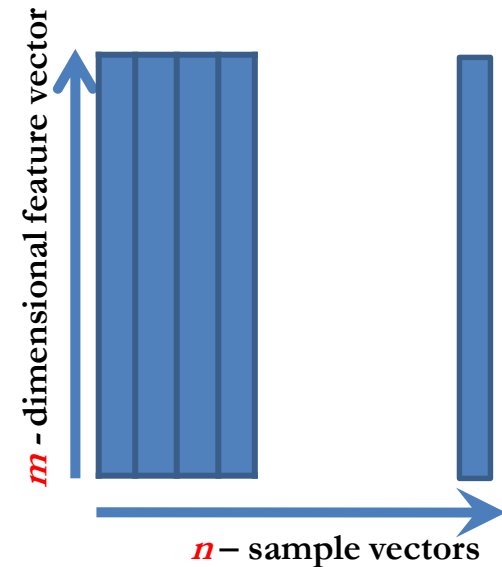
- LDA objective
- Recall ... PCA
- Now ... LDA
- LDA ... Two Classes
 - Counter example
- LDA ... C Classes
 - Illustrative Example
- LDA vs PCA Example
- Limitations of LDA

LDA Objective

- The objective of LDA is to perform dimensionality reduction ...
 - So what, PCA does this ...
- However, we want to preserve as much of the class discriminatory information as possible.
 - OK, that's new, let dwell deeper 😊 ...

Recall ... PCA

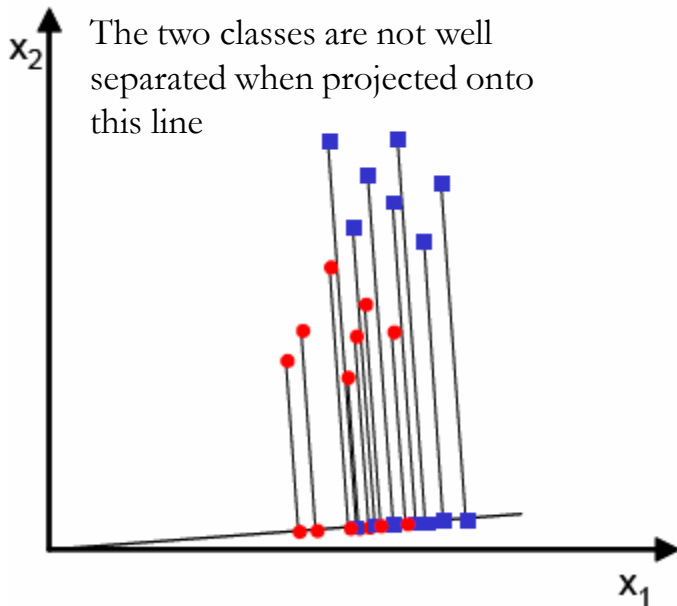
- In PCA, the main idea to re-express the available dataset to extract the relevant information by reducing the redundancy and minimize the noise.
- We didn't care about whether this dataset represent features from one or more classes, i.e. the discrimination power was not taken into consideration while we were talking about PCA.
- In PCA, we had a dataset matrix $\mathbf{X}$ with dimensions $m \times n$, where columns represent different data samples.
- We first started by subtracting the mean to have a zero mean dataset, then we computed the covariance matrix $\mathbf{S}_x = \mathbf{X}\mathbf{X}^T$.
- Eigen values and eigen vectors were then computed for $\mathbf{S}_x$. Hence the new basis vectors are those eigen vectors with highest eigen values, where the number of those vectors was our choice.
- Thus, using the new basis, we can project the dataset onto a less dimensional space with more powerful data representation.



Now ... LDA

- Consider a pattern classification problem, where we have C -classes, e.g. seabass, tuna, salmon ...
- Each class has $\mathbf{N}_i$ m -dimensional samples, where $i = 1, 2, \dots, C$.
- Hence we have a set of m -dimensional samples $\{\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^{\mathbf{N}_i}\}$ belong to class ω_i .
- Stacking these samples from different classes into one big fat matrix $\mathbf{X}$ such that each column represents one sample.
- We seek to obtain a transformation of $\mathbf{X}$ to $\mathbf{Y}$ through projecting the samples in $\mathbf{X}$ onto a hyperplane with dimension $C-1$.
- **Let's see what does this mean?**

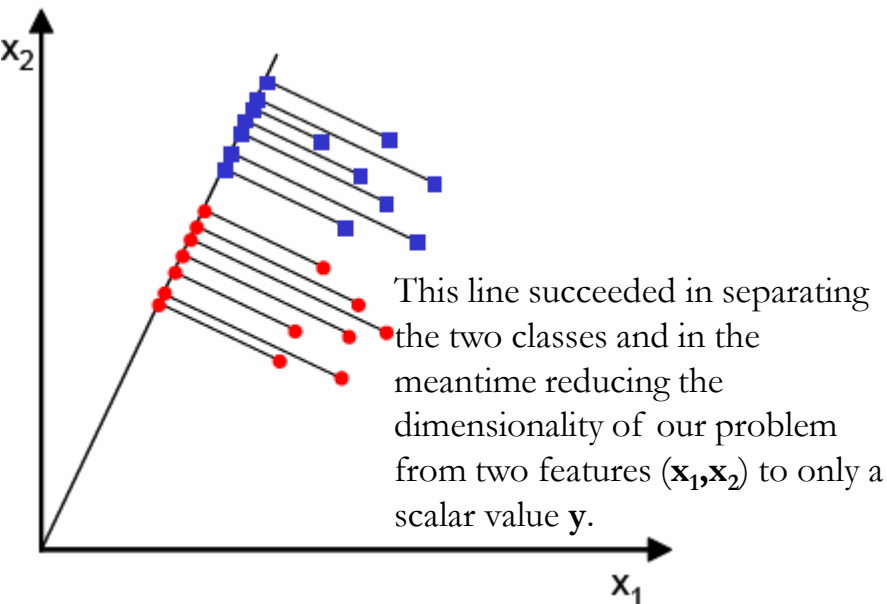
LDA ... Two Classes



Assume we have m -dimensional samples $\{\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N\}$, N_1 of which belong to ω_1 and N_2 belong to ω_2 .

We seek to obtain a scalar y by projecting the samples $\mathbf{x}$ onto a line (C-1 space, $C = 2$).

$$y = w^T x \quad \text{where} \quad x = \begin{bmatrix} x_1 \\ \cdot \\ \cdot \\ x_m \end{bmatrix} \quad \text{and} \quad w = \begin{bmatrix} w_1 \\ \cdot \\ \cdot \\ w_m \end{bmatrix}$$



Of all the possible lines we would like to select the one that maximizes the separability of the scalars.

- Mean

$$\mu_1 = W^T \mu_1$$

$$\mu_2 = W^T \mu_0$$

dimension of mean after projection = $d+1$

- Covariance

$$\Sigma_1 = W^T \Sigma_1 W$$

$$\Sigma_2 = W^T \Sigma_0 W$$

dimension of covariance after projection = $d \times d$

- Goal 1: Maximize the distance of projected mean

$$\Rightarrow (W^T \mu_1 - W^T \mu_0)^T (W^T \mu_1 - W^T \mu_0)$$

$$\Rightarrow (\mu_1 - \mu_0)^T W W^T (\mu_1 - \mu_0)$$

$$\Rightarrow W^T (\mu_1 - \mu_0) (\mu_1 - \mu_0)^T W$$

S_B = between class covariance

- $\max W^T S_B W$ objective function

Goal 2: For each class minimize the variance:

$$\Rightarrow w^T \leq_1 w + w^T \leq_0 w$$

$$\Rightarrow w^T (\leq_1 + \leq_0) w$$

S_w = within class covariance

$$- \boxed{\text{Min } w^T S_w w} \quad \text{objective func}$$

- Now, from Fisher linear discriminant,

$$\boxed{\text{max } \frac{w^T S_B w}{w^T S_w w}} \quad \text{objective func}$$

$$- \boxed{\begin{array}{l} \text{max } w^T S_B w \\ \text{subject to } w^T S_w w = 1 \end{array}} \rightarrow \text{objective func with constraint}$$

from Lagrange,

$$L(w, \lambda) = w^T S_B w - \lambda (w^T S_w w - 1)$$

Now, partial derivative,

$$\frac{\partial L}{\partial w} = 2S_B w - 2\lambda S_w w = 0$$

$$\Rightarrow S_B w = \lambda S_w w$$

Multiply both side by S_W^{-1}

$$\Rightarrow S_W^{-1} S_B w = \lambda w$$

Here, w is a eigenvector of $S_W^{-1} S_B$.

LDA ... Two Classes

- In order to find a good projection vector, we need to define a measure of separation between the projections.
- The mean vector of each class in $\mathbf{x}$ and $\mathbf{y}$ feature space is:

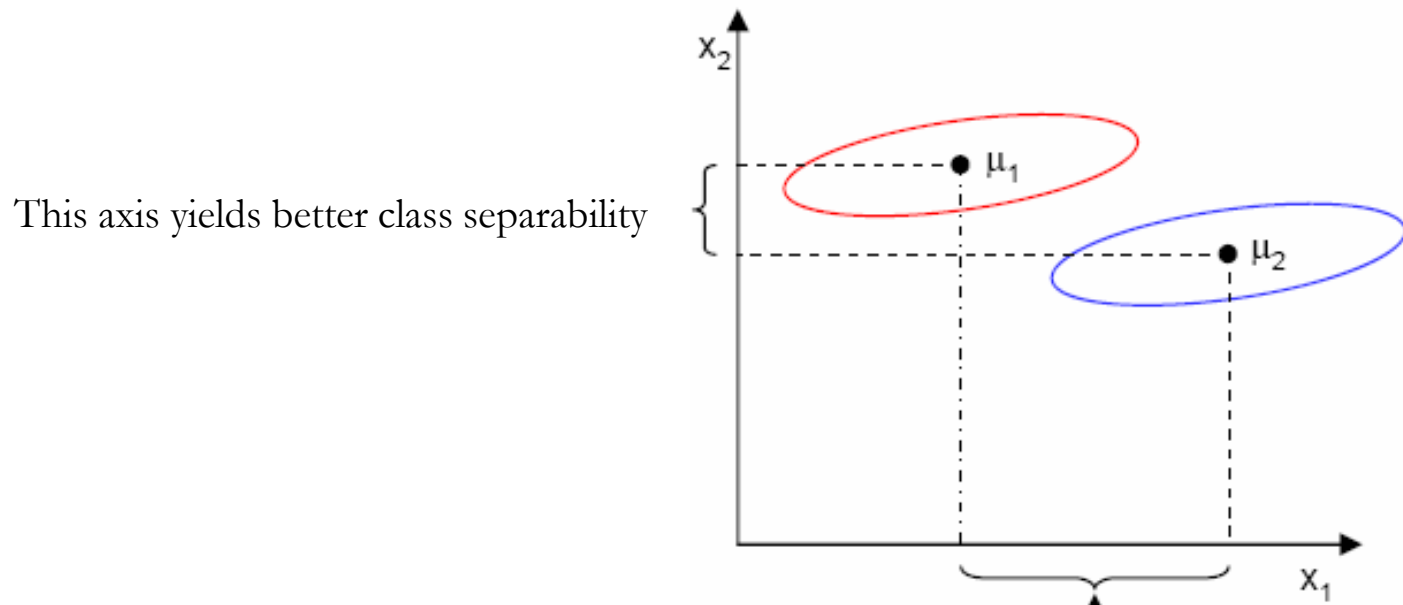
$$\mu_i = \frac{1}{N_i} \sum_{x \in \omega_i} x \quad \text{and} \quad \tilde{\mu}_i = \frac{1}{N_i} \sum_{y \in \omega_i} y = \frac{1}{N_i} \sum_{x \in \omega_i} w^T x$$
$$= w^T \frac{1}{N_i} \sum_{x \in \omega_i} x = w^T \mu_i$$

- We could then choose the distance between the projected means as our objective function

$$J(w) = \left| \tilde{\mu}_1 - \tilde{\mu}_2 \right| = \left| w^T \mu_1 - w^T \mu_2 \right| = \left| w^T (\mu_1 - \mu_2) \right|$$

LDA ... Two Classes

- However, the distance between the projected means is not a very good measure since it does not take into account the standard deviation within the classes.



This axis has a larger distance between means

LDA ... Two Classes

- The solution proposed by Fisher is to maximize a function that represents the difference between the means, normalized by a measure of the within-class variability, or the so-called *scatter*.
- For each class we define the scatter, an equivalent of the variance, as;

$$\tilde{s}_i^2 = \sum_{y \in \omega_i} (y - \tilde{\mu}_i)^2$$

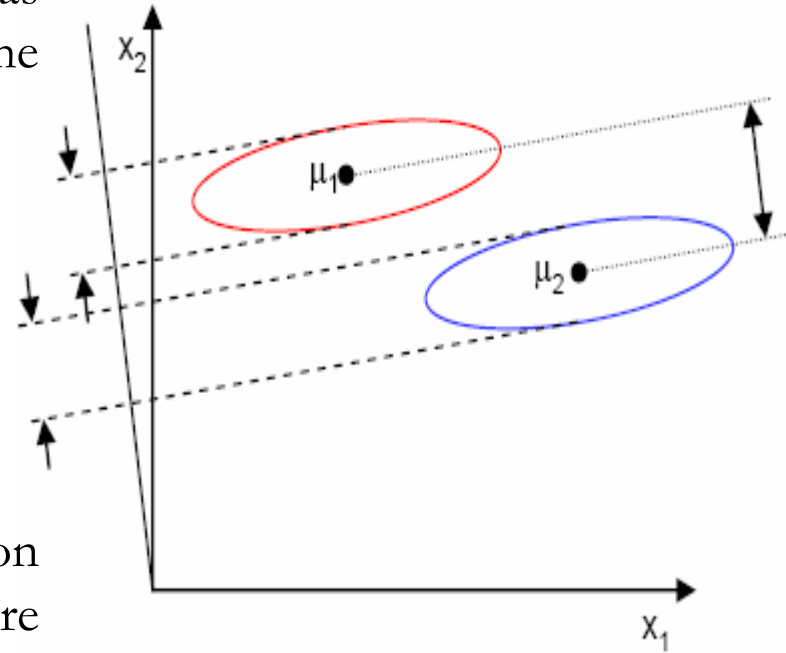
- $\tilde{s}_i^2$ measures the variability within class ω_i after projecting it on the y-space.
- Thus $\tilde{s}_1^2 + \tilde{s}_2^2$ measures the variability within the two classes at hand after projection, hence it is called *within-class scatter* of the projected samples.

LDA ... Two Classes

- The Fisher linear discriminant is defined as the linear function $\mathbf{w}^T \mathbf{x}$ that maximizes the criterion function:

$$J(\mathbf{w}) = \frac{|\tilde{\mu}_1 - \tilde{\mu}_2|^2}{\tilde{S}_1^2 + \tilde{S}_2^2}$$

- Therefore, we will be looking for a projection where examples from the same class are projected very close to each other and, at the same time, the projected means are as far apart as possible



LDA ... Two Classes

- In order to find the optimum projection $\mathbf{w}^*$, we need to express $J(\mathbf{w})$ as an explicit function of $\mathbf{w}$.
- We will define a measure of the scatter in multivariate feature space $\mathbf{x}$ which are denoted as *scatter matrices*;

$$S_i = \sum_{\mathbf{x} \in \omega_i} (\mathbf{x} - \mu_i)(\mathbf{x} - \mu_i)^T$$

$$S_w = S_1 + S_2$$

- Where $\mathbf{S}_i$ is the covariance matrix of class ω_i , and $\mathbf{S}_w$ is called the *within-class scatter matrix*.

LDA ... Two Classes

- Now, the scatter of the projection $\mathbf{y}$ can then be expressed as a function of the scatter matrix in feature space $\mathbf{x}$.

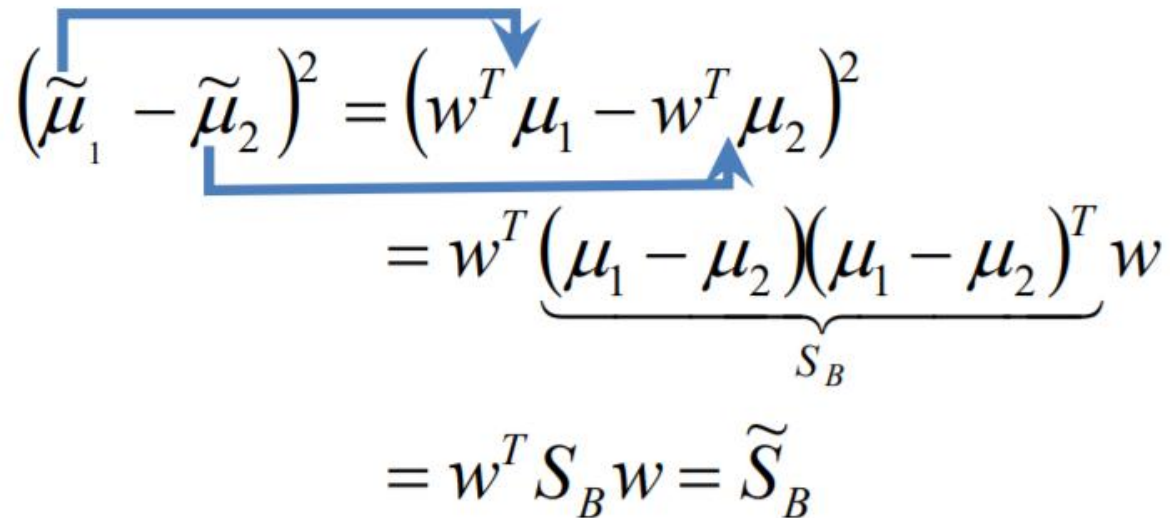
$$\begin{aligned}\tilde{S}_i^2 &= \sum_{y \in \omega_i} (y - \tilde{\mu}_i)^2 = \sum_{x \in \omega_i} (w^T x - w^T \mu_i)^2 \\ &= \sum_{x \in \omega_i} w^T (x - \mu_i)(x - \mu_i)^T w \\ &= w^T S_i w\end{aligned}$$

$$\tilde{S}_1^2 + \tilde{S}_2^2 = w^T S_1 w + w^T S_2 w = w^T (S_1 + S_2) w = w^T S_W w = \tilde{S}_W$$

Where $\tilde{S}_W$ is the within-class scatter matrix of the projected samples $\mathbf{y}$.

LDA ... Two Classes

- Similarly, the difference between the projected means (in y-space) can be expressed in terms of the means in the original feature space (x-space).


$$\begin{aligned}(\tilde{\mu}_1 - \tilde{\mu}_2)^2 &= (w^T \mu_1 - w^T \mu_2)^2 \\&= w^T \underbrace{(\mu_1 - \mu_2)(\mu_1 - \mu_2)^T}_{S_B} w \\&= w^T S_B w = \tilde{S}_B\end{aligned}$$

- The matrix $\mathbf{S}_B$ is called the *between-class scatter* of the original samples/feature vectors, while $\tilde{\mathbf{S}}_B$ is the between-class scatter of the projected samples $\mathbf{y}$.
- Since $\mathbf{S}_B$ is the outer product of two vectors, its rank is at most one.

LDA ... Two Classes

- We can finally express the Fisher criterion in terms of S_W and S_B as:

$$J(w) = \frac{|\tilde{\mu}_1 - \tilde{\mu}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2} = \frac{w^T S_B w}{w^T S_W w}$$

- Hence $J(w)$ is a measure of the difference between class means (encoded in the between-class scatter matrix) normalized by a measure of the within-class scatter matrix.

LDA ... Two Classes

- To find the maximum of $J(w)$, we differentiate and equate to zero.

$$\begin{aligned}\frac{d}{dw} J(w) &= \frac{d}{dw} \left(\frac{w^T S_B w}{w^T S_W w} \right) = 0 \\ \Rightarrow (w^T S_W w) \frac{d}{dw} (w^T S_B w) - (w^T S_B w) \frac{d}{dw} (w^T S_W w) &= 0 \\ \Rightarrow (w^T S_W w) 2S_B w - (w^T S_B w) 2S_W w &= 0\end{aligned}$$

Dividing by $2w^T S_W w$:

$$\begin{aligned}\Rightarrow \left(\frac{w^T S_W w}{w^T S_W w} \right) S_B w - \left(\frac{w^T S_B w}{w^T S_W w} \right) S_W w &= 0 \\ \Rightarrow S_B w - J(w) S_W w &= 0 \\ \Rightarrow S_W^{-1} S_B w - J(w) w &= 0\end{aligned}$$

LDA ... Two Classes

- Solving the generalized eigen value problem

$$S_W^{-1} S_B w = \lambda w \quad \text{where} \quad \lambda = J(w) = \text{scalar}$$

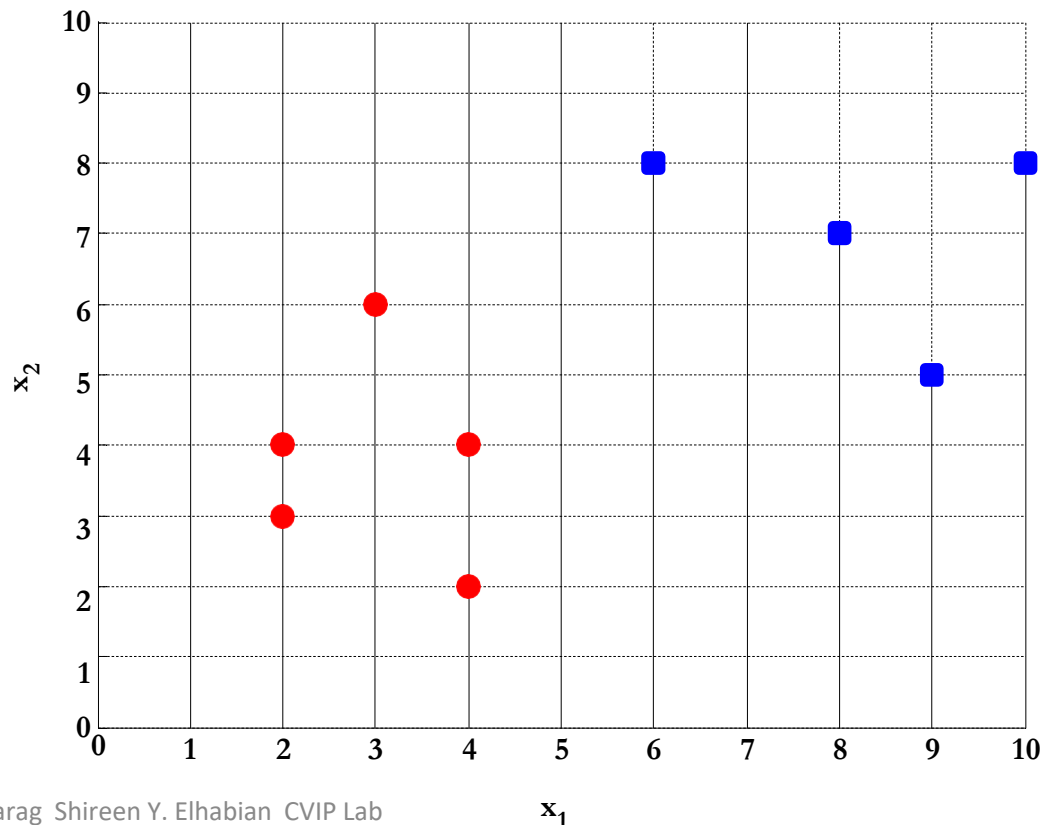
yields

$$w^* = \arg \max_w J(w) = \arg \max_w \left(\frac{w^T S_B w}{w^T S_W w} \right) = S_W^{-1} (\mu_1 - \mu_2)$$

- This is known as Fisher's Linear Discriminant, although it is not a discriminant but rather a specific choice of direction for the projection of the data down to one dimension.
- Using the same notation as PCA, the solution will be the eigen vector(s) of $S_X = S_W^{-1} S_B$

LDA ... Two Classes - Example

- Compute the Linear Discriminant projection for the following two-dimensional dataset.
 - Samples for class ω_1 : $\mathbf{X}_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$
 - Sample for class ω_2 : $\mathbf{X}_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$



```
% samples for class 1
X1 = [4,2;
      2,4;
      2,3;
      3,6;
      4,4];

% samples for class 2
X2 = [9,10;
      6,8;
      9,5;
      8,7;
      10,8];
```

LDA ... Two Classes - Example

- The classes mean are :

$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x = \frac{1}{5} \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \end{pmatrix} \right] = \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}$$

$$\mu_2 = \frac{1}{N_2} \sum_{x \in \omega_2} x = \frac{1}{5} \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} 9 \\ 5 \end{pmatrix} + \begin{pmatrix} 8 \\ 7 \end{pmatrix} + \begin{pmatrix} 10 \\ 8 \end{pmatrix} \right] = \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix}$$

```
% class means  
Mu1 = mean(X1) ' ;  
Mu2 = mean(X2) ' ;
```

LDA ... Two Classes - Example

- Covariance matrix of the first class:

$$\begin{aligned} S_1 &= \sum_{x \in \omega_1} (x - \mu_1)(x - \mu_1)^T = \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^2 \\ &\quad + \left[\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^2 \\ &= \begin{pmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} \end{aligned}$$

```
% covariance matrix of the first class  
S1 = cov(X1);
```

LDA ... Two Classes - Example

- Covariance matrix of the second class:

$$\begin{aligned} S_2 &= \sum_{x \in \omega_2} (x - \mu_2)(x - \mu_2)^T = \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^2 \\ &\quad + \left[\begin{pmatrix} 9 \\ 5 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 8 \\ 7 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^2 + \left[\begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^2 \\ &= \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix} \end{aligned}$$

```
% covariance matrix of the first class  
S2 = cov(X2);
```

LDA ... Two Classes - Example

- Within-class scatter matrix:

$$\begin{aligned} S_w = S_1 + S_2 &= \begin{pmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} + \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix} \\ &= \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix} \end{aligned}$$

```
% within-class scatter matrix  
Sw = S1 + S2 ;
```


LDA ... Two Classes - Example

- Between-class scatter matrix:

$$\begin{aligned} S_B &= (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \\ &= \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T \\ &= \begin{pmatrix} -5.4 \\ -3.8 \end{pmatrix} \begin{pmatrix} -5.4 & -3.8 \end{pmatrix} \\ &= \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix} \end{aligned}$$

```
% between-class scatter matrix  
SB = (Mu1-Mu2) * (Mu1-Mu2) ' ;
```

LDA ... Two Classes - Example

- The LDA projection is then obtained as the solution of the generalized eigen value problem

$$S_W^{-1}S_B w = \lambda w$$

$$\Rightarrow |S_W^{-1}S_B - \lambda I| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix}^{-1} \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{pmatrix} \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 9.2213 - \lambda & 6.489 \\ 4.2339 & 2.9794 - \lambda \end{pmatrix} \right|$$

$$= (9.2213 - \lambda)(2.9794 - \lambda) - 6.489 \times 4.2339 = 0$$

$$\Rightarrow \lambda^2 - 12.2007\lambda = 0 \Rightarrow \lambda(\lambda - 12.2007) = 0$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = 12.2007$$

LDA ... Two Classes - Example

- Hence

$$\begin{pmatrix} 9.2213 & 6.489 \\ 4.2339 & 2.9794 \end{pmatrix} w_1 = \underbrace{0}_{\lambda_1} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

and

$$\begin{pmatrix} 9.2213 & 6.489 \\ 4.2339 & 2.9794 \end{pmatrix} w_2 = \underbrace{12.2007}_{\lambda_2} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

Thus;

$$w_1 = \begin{pmatrix} -0.5755 \\ 0.8178 \end{pmatrix} \quad \text{and} \quad w_2 = \begin{pmatrix} 0.9088 \\ 0.4173 \end{pmatrix} = w^*$$

```
% computing the LDA projection
invSw = inv(Sw);

invSw_by_SB = invSw * SB;

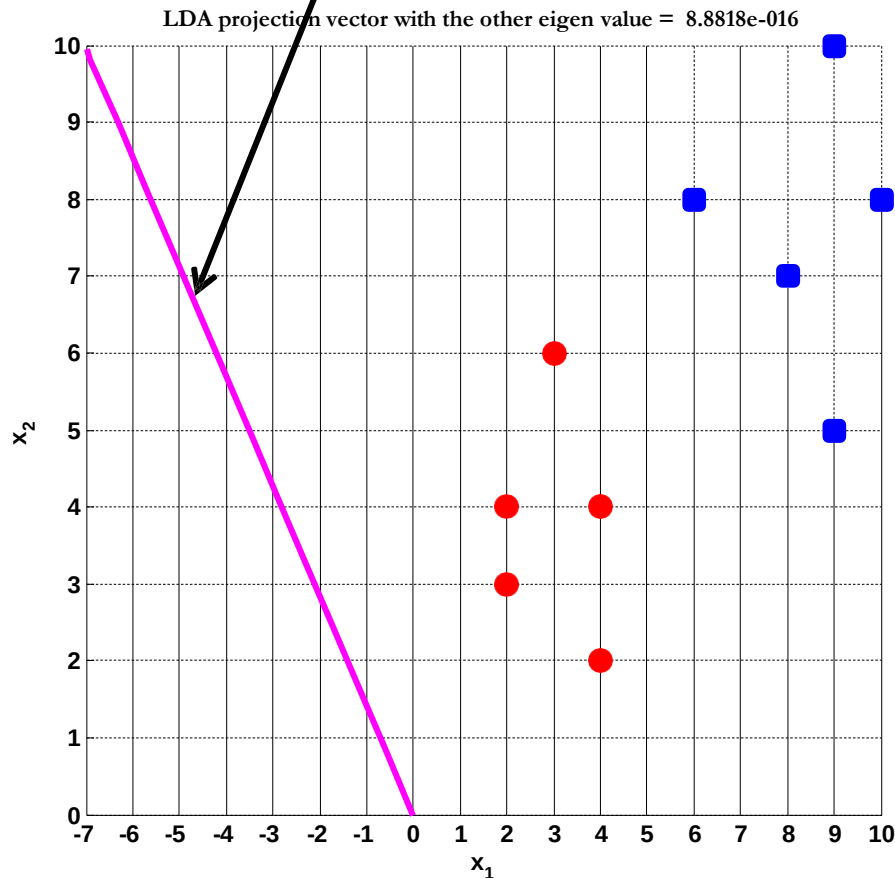
% getting the projection vector
[V,D] = eig(invSw_by_SB)

% the projection vector
W = V(:,1);
```

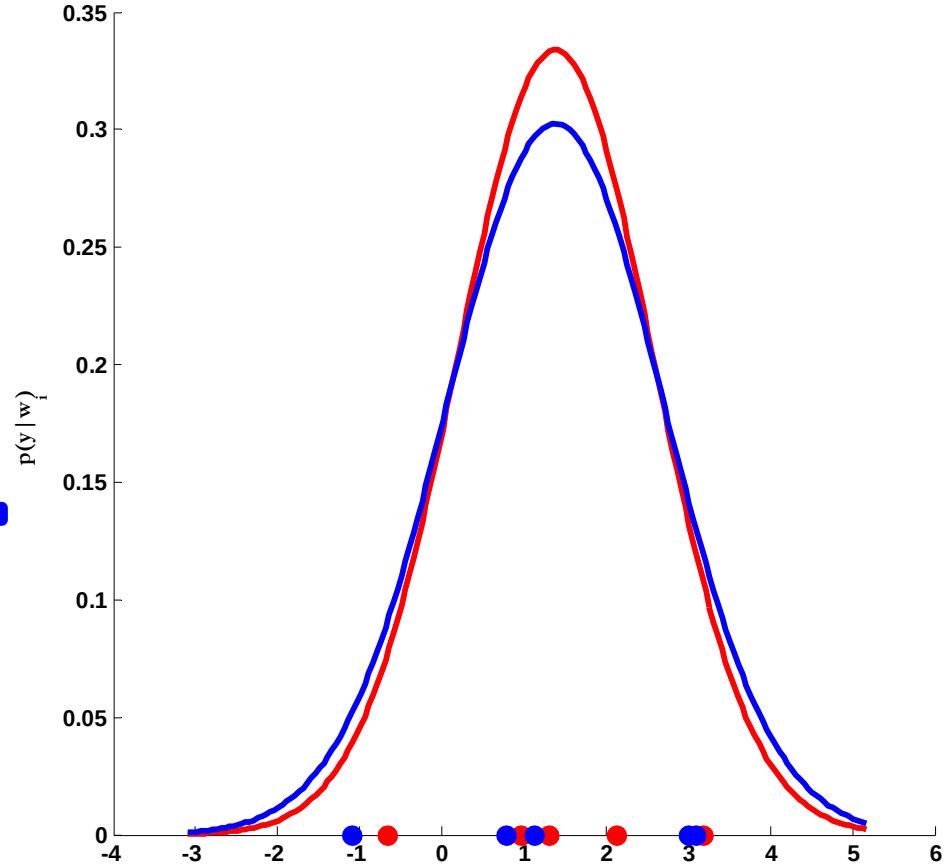
- The optimal projection is the one that given maximum $\lambda = J(w)$

LDA - Projection

The projection vector corresponding to the **smallest** eigen value



Classes PDF : using the LDA projection vector with the other eigen value = 8.8818e-016

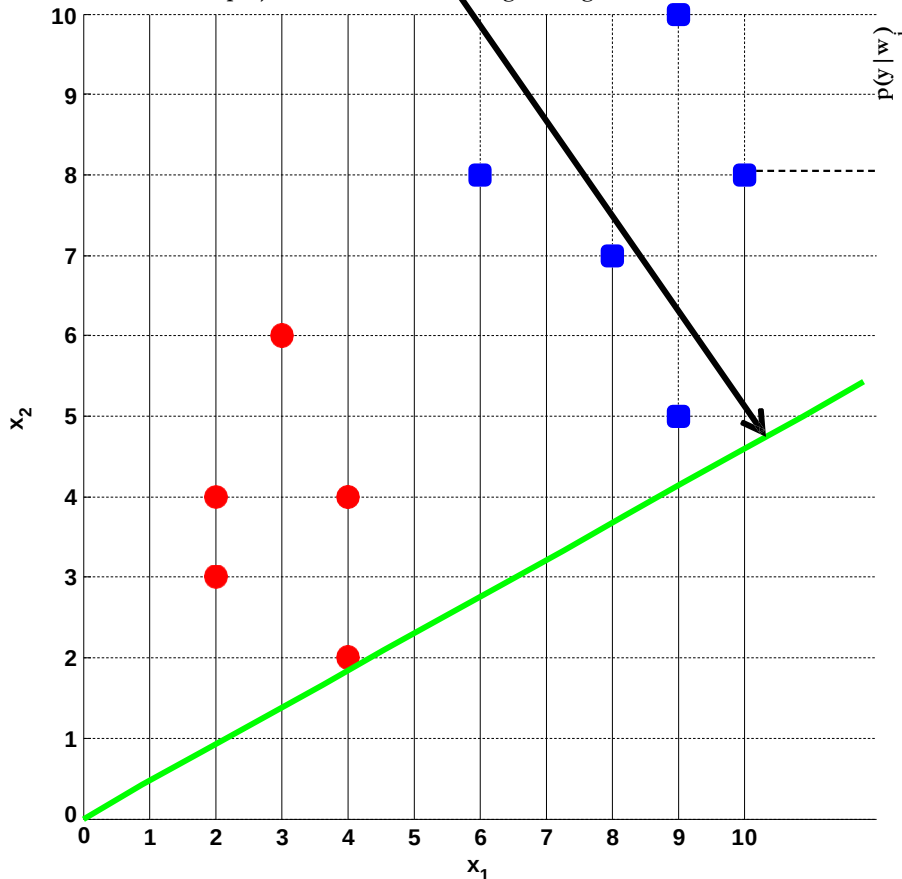


Using this vector leads to **bad separability** between the two classes

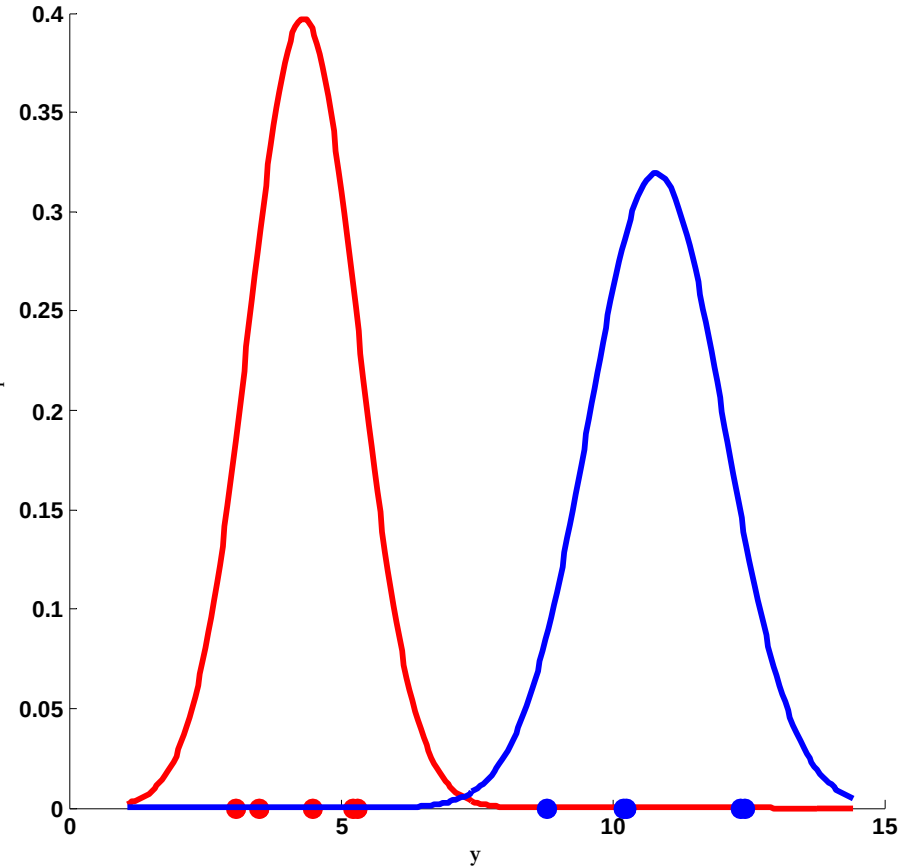
LDA - Projection

The projection vector corresponding to the **highest** eigen value

LDA projection vector with the highest eigen value = 12.2007

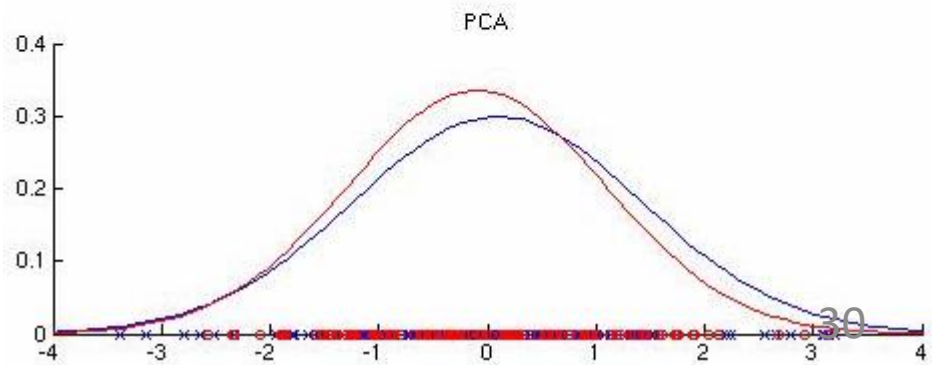
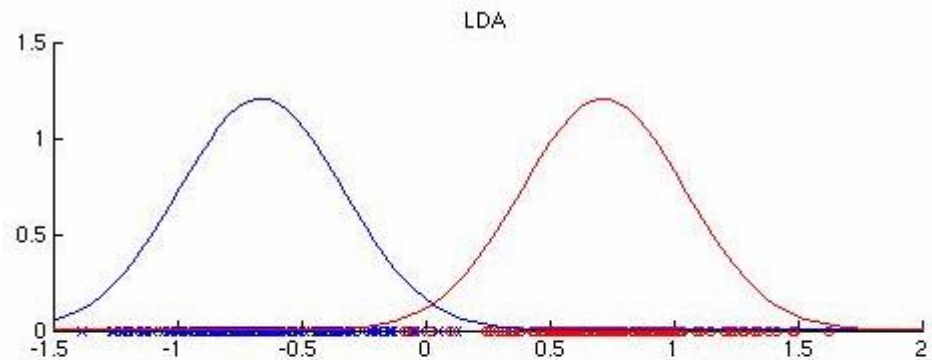
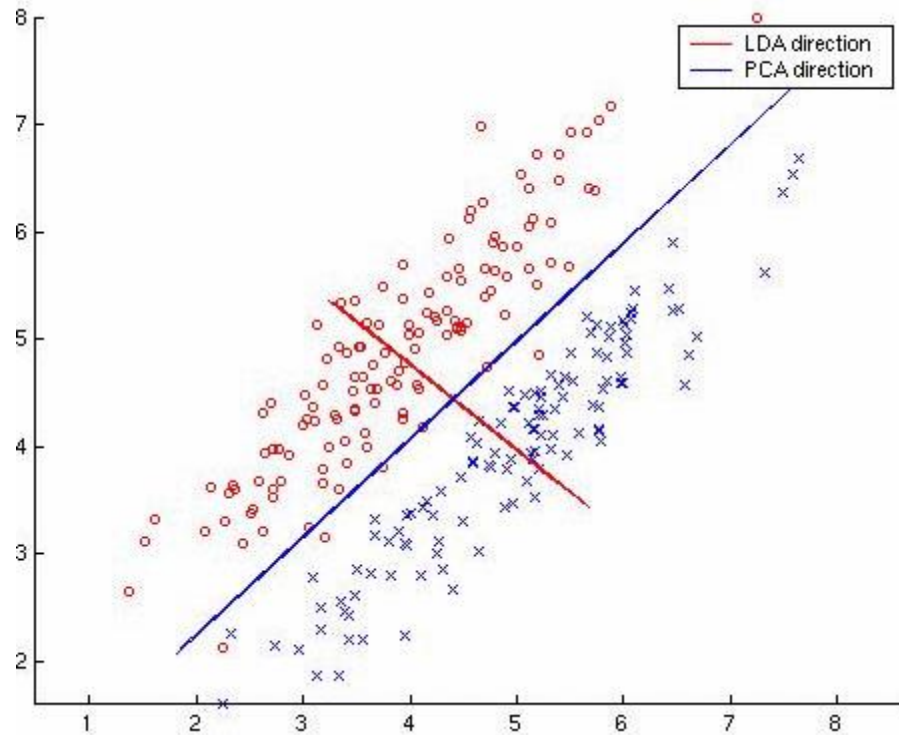


Classes PDF : using the LDA projection vector with highest eigen value = 12.2007



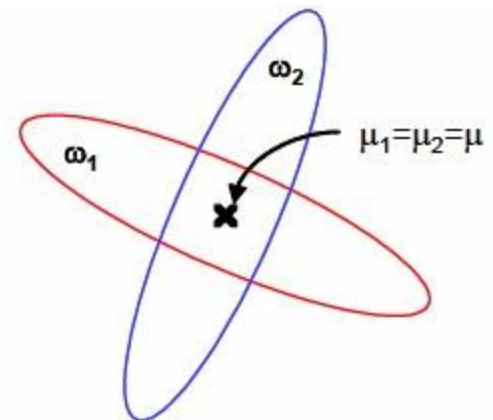
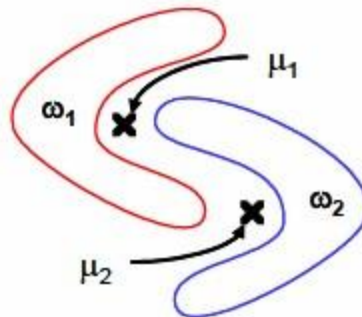
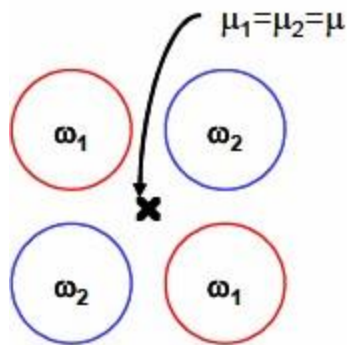
Using this vector leads to **good separability** between the two classes

PCA vs LDA



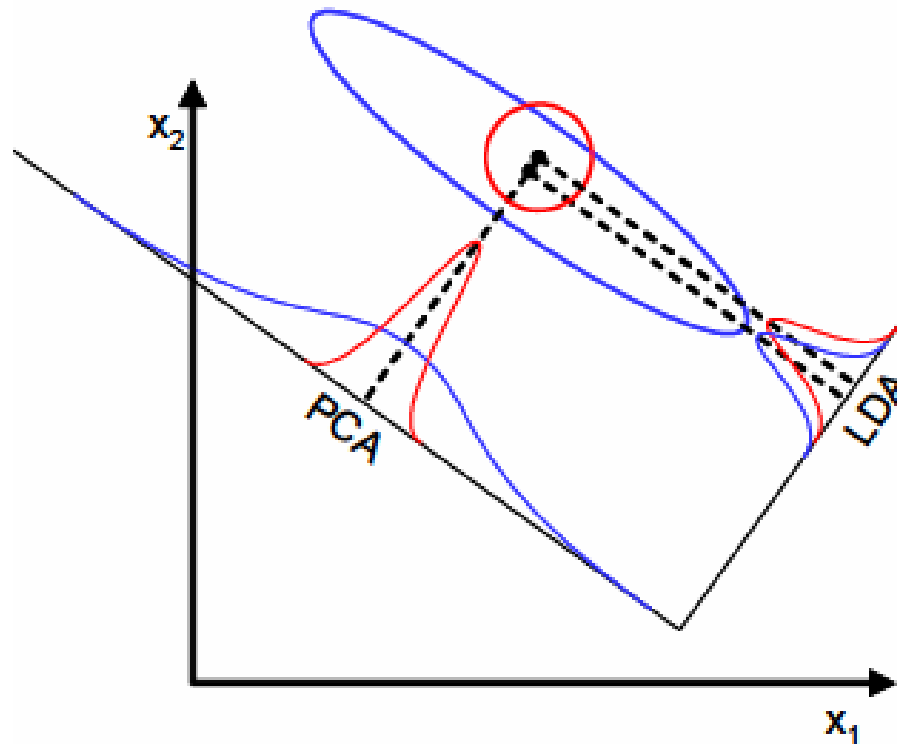
Limitations of LDA

- **LDA produces at most $C-1$ feature projections**
 - If the classification error estimates establish that more features are needed, some other method must be employed to provide those additional features
- **LDA is a parametric method since it assumes unimodal Gaussian likelihoods**
 - If the distributions are significantly non-Gaussian, the LDA projections will not be able to preserve any complex structure of the data, which may be needed for classification.



Limitations of LDA

- LDA will fail when the discriminatory information is not in the mean but rather in the variance of the data



Thank You